Chapter 16 · 09

Docker Without Kubernetes

Docker Swarm · Portainer · Standalone Production Patterns

Analogy

• Not every app needs Kubernetes — that's like hiring a city traffic control system for a single delivery van. Docker Swarm is like a small dispatch office: simple routing, basic scaling, no Kubernetes complexity. For many small/medium apps, Docker Compose on a single VM or Swarm on 3 nodes is perfectly sufficient.

Key Terms

Docker Swarm	Docker's built-in orchestration. Turns multiple Docker hosts into a cluster. Simpler than K8s
Manager node	Swarm node that maintains cluster state, schedules services. Should be 3 or 5 for HA
Worker node	Runs containers assigned by manager. No scheduling authority
Service (Swarm)	docker service create — desired state declaration. Swarm maintains replica count
Stack (Swarm)	docker stack deploy — deploys a Compose file to a Swarm cluster
Overlay network	Swarm's default network driver — connects containers across nodes transparently
Rolling update	docker service update --update-parallelism 1 — replaces replicas one at a time
Portainer	Web UI for managing Docker (standalone or Swarm). RBAC, stacks, container management
Watchtower	Auto-updates running containers when new images are pushed to the registry
Traefik	Reverse proxy / ingress for Docker. Auto-discovers containers via labels, handles TLS

Docker Swarm vs Kubernetes

Feature	Docker Swarm	Kubernetes
Learning curve	Low — uses Compose syntax	High — many new concepts
HA control plane	3 manager nodes	etcd + control plane components
Auto-scaling	Manual / Swarm service scale	HPA, KEDA, Cluster Autoscaler
Rolling updates	Built-in, basic	Deployment strategies + Argo Rollouts
Ecosystem	Smaller	Massive — Helm, operators, service mesh

Best for	Small teams, simple apps	Large scale, complex workloads

Docker Swarm Quick Start

```
# Initialize Swarm on first node
docker swarm init --advertise-addr 192.168.1.10
# Join worker nodes (token from above command)
docker swarm join --token SWMTKN-1-xxx 192.168.1.10:2377
# Deploy a stack from docker-compose.yml
docker stack deploy -c docker-compose.yml shopverse
# Scale app service to 3 replicas
docker service scale shopverse_app=3
# Rolling update with zero downtime
docker service update \
  --image myapp:2.0.0 \
  --update-parallelism 1 \
  --update-delay 10s \
  shopverse_app
# List services
docker service ls
docker service ps shopverse_app
```

■ Real-World: Basecamp

• Basecamp (Hey.com) runs their production infrastructure on Docker without Kubernetes. A small fleet of servers runs Docker Compose/Swarm with Traefik as the load balancer. They've written publicly about rejecting Kubernetes complexity — their 10-person ops team manages millions of users on straightforward Docker infrastructure.